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RESEARCH INTERESTS

Natural language processing, with a focus on **measurement validity** — whether the benchmarks and automatic judges used to rank models measure what they claim to. One result recurs across my projects: **surface competence $\neq$ underlying competence**. I want to measure that gap, explain it mechanistically, and close it.

EDUCATION

Northeastern University, China — B.Eng., Computer Science and Technology 2023 – 2027 (expected)

- CGPA 4.05 / 5.00 (90 / 100) · Rank 5 / 100 in major cohort
- Advised by Shiqi Zhao (Assistant Professor, Northeastern University) since 2024

PUBLICATIONS

[1] **Beyond Survival: Evaluating LLMs in Social Deduction Games with Human-Aligned Strategies**

Zirui Song*, Yuan Huang*, Junchang Liu*, et al. (**equal contribution*)

EMNLP 2026, Main Conference · co-first author · arXiv:2510.11389 · Dataset

[2] **Geolocation with Real Human Gameplay Data: A Large-Scale Dataset and Human-Like Reasoning Framework**

Zirui Song, Jingpu Yang, Yuan Huang, et al.

Preprint · arXiv:2502.13759 · Dataset · Code

RESEARCH EXPERIENCE

EditJudgeBias — counterfactual audit of multimodal image-editing judges 2026 – present

Sole first author · in preparation, targeting ICLR 2027

- **Design.** Admit only perturbations that can be shown not to have changed edit quality: preservation verification is an admission gate, not a post-hoc check, and every cue carries a zero-dose sham control, so null results are readable.
- **Scale.** 13 cue conditions across four injection sites, 5 published judges plus an independent replication arm — 97,973 judgments, 0.300% parse failure.
- **Findings.** Reordering the two candidates alone flips decisions at 38.2× the judge’s self-noise; of three mitigation layers only the protocol-level one works, and it works by **abstention, not debiasing**.
- **Rigor.** 1,307 unit tests and 109 sha256-stamped frozen artifacts; the audit overturned four of my own conclusions.

Co-first author · inception through submission

- Benchmark built from 100+ hours of professional human social-deduction gameplay (32.4M utterance tokens, 30 roles, 15 rule variants), scored against winning-side MVP trajectories rather than LLM self-play; 5 verbal dimensions and 2 decision tasks.
- Half of the evaluated models score below 0.50 (best 0.720). Deception reasoning and counterfactual trade-off are weakest while persuasive generation stays strong — fluency ≠ strategic correctness.

GeoComp / GeoCoT / GeoEval — geolocation from real human gameplay data

Preprint

Co-author (3 of 9)

- Dataset from a public geo-guessing platform — 740K users, 2.7M locations, 25M human response records — with human accuracy calibrating item difficulty. Multi-step geographic reasoning improved standard metrics by up to 25%; scoring the reasoning process rather than only the final answer added a further 9%.

HONORS & AWARDS

- China National Scholarship — national level · 2023–2024
- Northeastern University Comprehensive Scholarship · 2024–2026
- Third Prize, National Finals, Lanqiao Cup Programming Competition · 2024

SKILLS

- Python, C++, Java · PyTorch, Hugging Face Transformers · backend engineering intern, 2024 (Java / Spring Boot)